



mowito
robotics

Wire Harness Automation

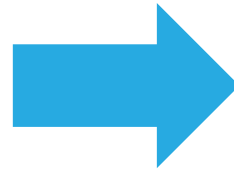
Mowito.ai

Mowito-NeuralStack

Multimodal AI model that learns specific from demonstrations. **Reconfigurable on the fly - by factory workers**

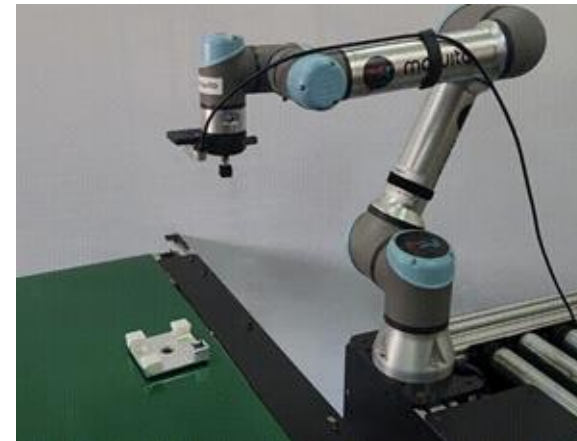
1 Teach

The Physical AI model learns from Demonstration of Task



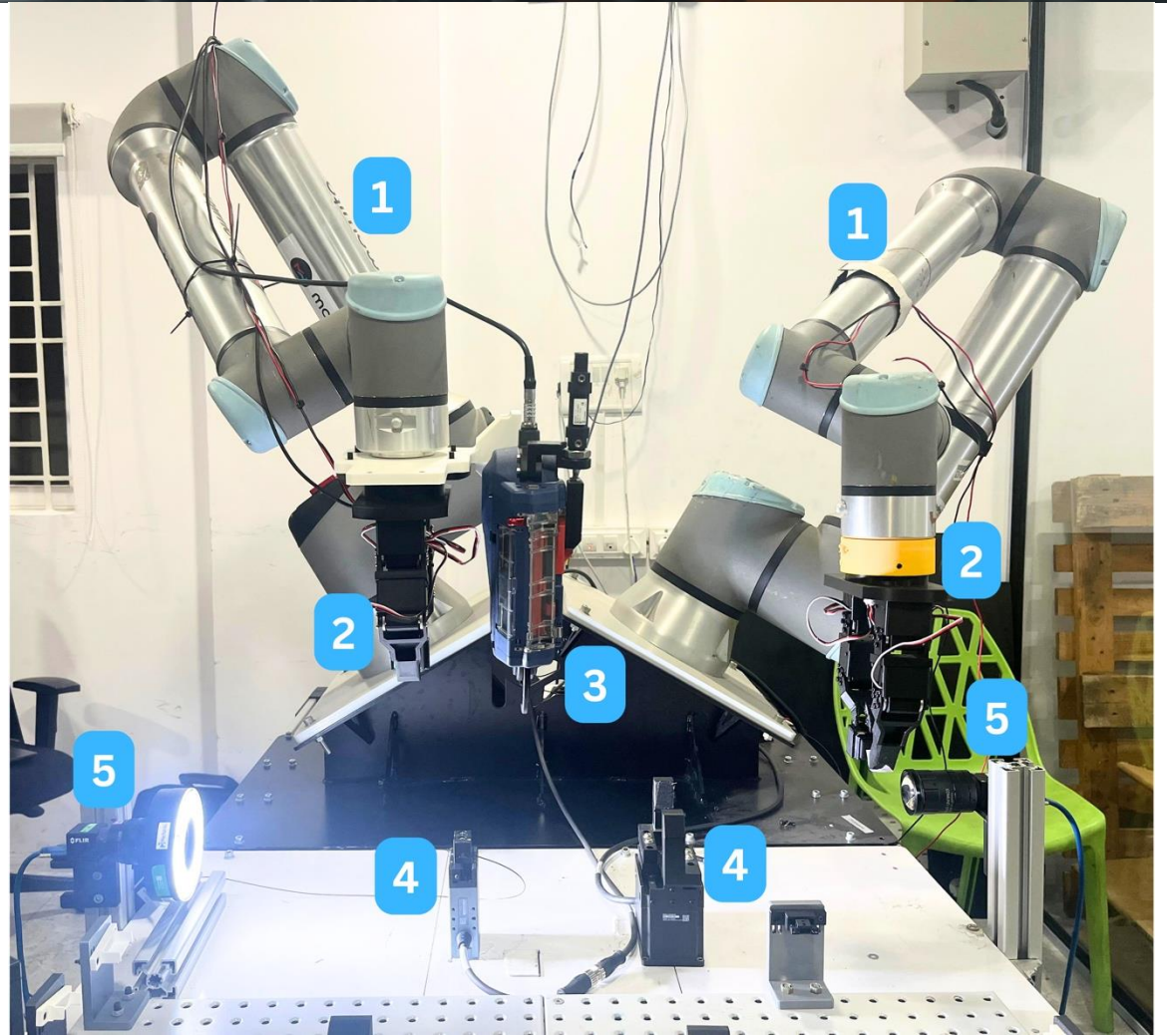
2 Deploy

The Robot will dynamically generate the right path



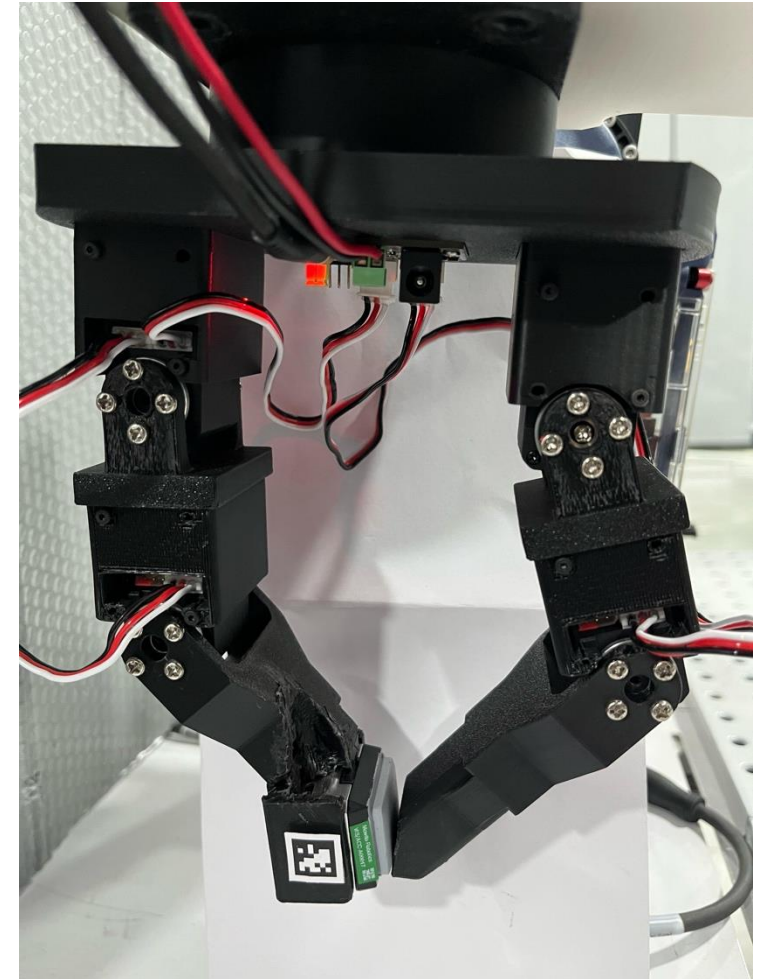
System configuration

- 1) Dual arm system with custom gripper - NeuralGripper
- 2) NeuralGripper
 - a. 4-Dof 2 finger gripper
 - b. Tactile sensor at one end
- 3) Hellerman-Tyton automatic clamping tool
- 4) Static 2 finger grippers
- 5) Static 2D cameras



System configuration

1. Each gripper has two fingers, and each finger has 2 degrees of freedoms.
2. Further, at the tip of each finger, we have a tactile sensor.
3. On the left gripper we have Hellermann-Tyton's clamping tool.



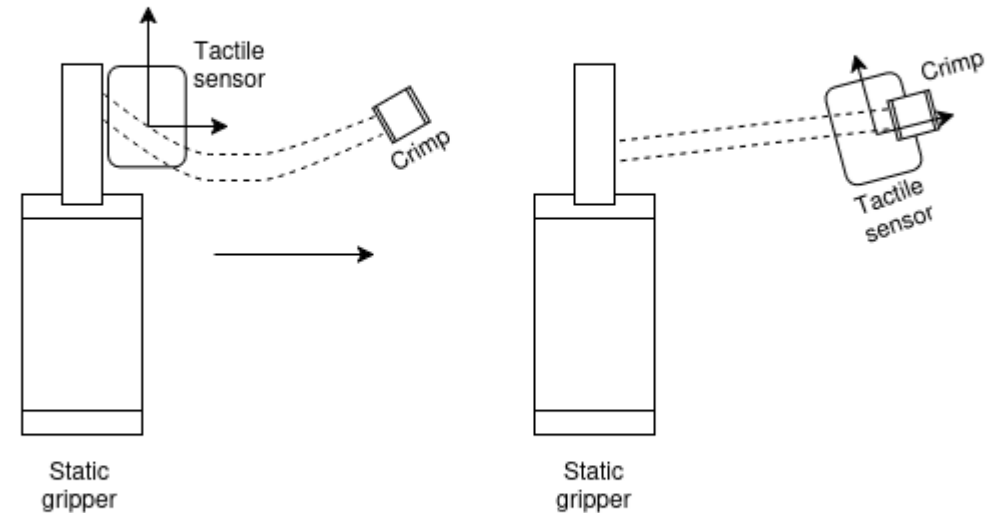
Wire-end localisation

Using tactile feedback and static 2D camera input to the model, the robotic arm is able to align with a constant relative pose of the crimp with respect to the arm end-effector



Wire-end localisation

The gripper-tactile-tip goes to the beginning of fixed static gripper, follows along the wire with varying gripping force so as to maintain partial grip and good tactile signature



Wire-end localisation



In-hand manipulation of wire

Neuralgripper allows pinch-and-roll control policy to achieve correct orientation needed, this can be used for large error in orientation (>60 deg).





Primary challenge considered

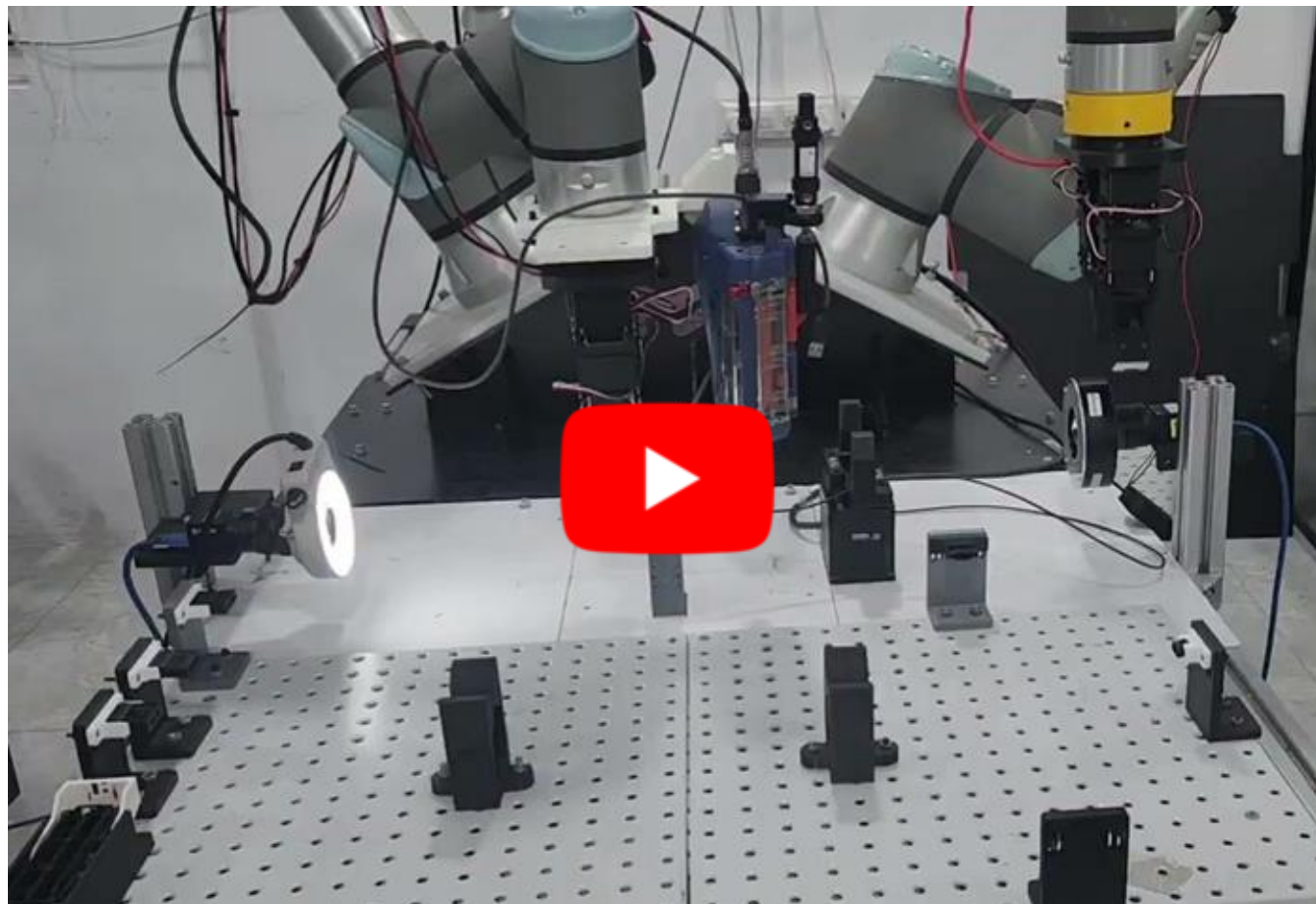
- Continuous localization of wire end
- Pose informed in-hand manipulation of flexible wires



Assembly steps

1. Load single wire onto the 2 static gripper
2. Dual arm with NeuralGripper localise the crimp on each end
3. If the crimp needs rotational adjustment in the axis of the wire, NeuralGripper does in-hand manipulation
4. Through tactile feedback, both crimp-ends are localised
5. Crimp ends are inserted to connectors , whose pose are known
6. Connectors with secondary lock is pressed
7. Connectors needing to be placed in the housing is done so through tactile feedback
8. Automatic camping tool is used to tie the zip tie at 50mm interval

Final Video





Current challenges

1. Lack of repeatability from neuralgripper mechanism
 - a. Backlash from geared motor
 - b. Shrinkage from 3D printing process cause manufacturing discrepancies
2. Lack of variation in dataset
 - a. More variability in dataset, particularly in the feedback from tactile sensor



Economics of the solution

- The entire solution-Hardware and software combined can come to a cost of €120.000.
- The current hardware can also be extended to do singulation of wire from a heaped bin, this will affect cycle time, but a separate constrained singulation mechanism is not needed

Thank You!

